25901408 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Value Addition Course (VAC) - MG4VACMBG200 - SANITATION MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Notes Questions

Answer any **10** questions

Each question carries **2** marks

1. Give two reasons why protective clothing is necessary in fermentation units. [U] / [CO1]
2. Define animal inoculation. [K] / [CO1]
3. What is a HEPA filter? [K] / [CO1]
4. Define droplet nuclei? [U] / [CO2]
5. Define air sampling devices? [U] / [CO2]
6. What are the preventive measures of diphtheria? [U] / [CO2]
7. Name two common bacteria found in sewage. [K] / [CO3]
8. Define primary treatment. [K] / [CO3]
9. Mention the role of methanogens in anaerobic treatment. [K] / [CO3]

10. Define aeration tanks of water pollution. [K] / [CO3]

11. Name viral waterborne diseases. [K] / [CO3]

12. Mention the benefits of composting for soil fertility. [K] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. What standard precautions are followed in hospitals and laboratories? [K] / [CO1]

14. How does proper sanitation help in preventing food contamination? [K] / [CO1]

15. Explain the factors affecting the survival of micro organisms in air? [U] / [CO2]

16. Explain the impaction method of air sampling? [U] / [CO2]

17. Explain why nitrogen and phosphorus need to be removed. [U] / [CO3]

18. Explain shortly why proper disposal of agricultural waste is necessary. [U] / [CO4]

[5x4 = 20]

Part C

Essay Question

Answer any **1** questions

Each question carries **10** marks

19. Discuss the applications and importance of air sanitation techniques in public places, hospitals and industries? [A] / [CO2]

20. Explain the causes, symptoms, transmission, and preventive measures of any two parasitic waterborne diseases. [U] / [CO3]

[10x1 = 10]